ITECH2002 Week 3 Tutorial

Introduction

Participate in the class discussion and then complete the individual tasks.
Record your work in a document for later reference.

Prerequisites

You should have completed all the learning materials for week 3 provided in the "Week 3 Learning Book" on Moodle – watch all the videos included in the Book.

If you feel unsure about any of the topics, you should make sure you have read Chapter 3 from the textbook (available digitally through the library – see link on Moodle) up to page 58.

Motivation

This tutorial helps you in regards to the following learning outcomes:

1. Identify use cases for a given scenario.
2. Use the user-goal technique to identify use cases.
3. Use the event decomposition technique to identify use cases.
4. Write use cases as user stories (or write user stories as use cases)

Task 1: Class Discussion

The tutor will lead a class discussion based around the following questions:

1. What is meant when we talk about a "use case" in relation to an information system?
2. What is meant by "actor" in relation to a use case?
3. True or false: All systems only have one actor that we need to consider.
4. What is the difference between a use case name, and a use case (short) description?
5. What is meant by "User story"?
6. What are the three typical parts (phrases) included in all user stories?

7. Review the external event checklist in Figure 3-4 in the textbook (or see below) and then think about a university course registration system (to keep records of student enrolment and results data for all semesters). What is an example of an event of each type in the checklist? Name each event by using the guidelines for naming an external event.

External events to look for include:

√ External agent wants something resulting in a transaction
√ External agent wants some information
√ Data changed and needs to be updated
√ Management wants some information

Figure 3-4. External Event Checklist

8. Review the temporal event checklist in Figure 3-5 in the textbook (or see below). Would a student grade report be an internal or external output? Would a class list for the instructor be an internal or external output? What are some other internal and external outputs for a course registration system? Using the guidelines for naming temporal events, what would you name the events that trigger these outputs?

Temporal events to look for include:

√ Internal outputs needed
  √ Management reports (summary or exception)
  √ Operational reports (detailed transactions)
  √ Internal statements and documents (including payroll)
√ External outputs needed
  √ Statements, status reports, bills, reminders

Figure 3-5. Temporal Event Checklist

9. Consider the following sequence of actions taken by a customer at a bank. Which action is the event the analyst should define for a bank account transaction processing system?
   (1) Kevin gets a check from Grandma for his birthday.
   (2) Kevin wants a car.
   (3) Kevin decides to save his money.
   (4) Kevin goes to the bank.
   (5) Kevin waits in line.
   (6) Kevin makes a deposit in his savings account.
   (7) Kevin grabs the deposit receipt.
   (8) Kevin asks for a brochure on auto loans.

10. The perfect technology assumption states that use cases should be included during analysis only if the system would be required to respond under perfect conditions. Why?

Task 2: Individual work

1. Visit some websites of car manufacturers (such as Honda, BMW, Toyota, and Acura). Many of these sites have a use case that is typically named *Build and price a car* (or similar). As a potential customer, you can select a car model, select features and options, and get the car's suggested price and list of specifications.

   Write a brief use case description for this use case.

2. Again looking at a website for one of the car manufacturers, consider yourself a potential buyer and then identify all the use cases included on the site that correspond to your goals.

Task 3

Refer to the RMO CSMS Order Fulfillment subsystem shown in Figure 3-10b of the textbook (or see below). Draw on a piece of paper: a use case diagram that shows all actors and all use cases for that subsystem.

CSMS Order Fulfillment Subsystem	
Use cases	**Users/actors**
Ship items	Shipping
Manage shippers	Shipping
Create backorder	Shipping
Create item return	Shipping, customer
Look up order status	Shipping, customer, management
Track shipment	Shipping, customer, marketing
Rate and comment on product	Customer
Provide suggestion	Customer
Review suggestion	Management

Figure 3-10b: Use cases and actors for CSMS Sales Subsystems

Task 4 (Case Study)

ON THE SPOT COURIER SERVICE (CASE STUDY FROM TEXTBOOK, CHAPTER 3)

Recall the *On the Spot courier service* introduced in Chapter 2 (Week 2 Tutorial case study). Further details are now provided, in particular, the details of the package pickup and delivery process are described here:

When Bill got an order, only on his phone at first, he recorded when he received the call and when the shipment would be ready for pickup. Sometimes, customers wanted immediate pickup; sometimes, they were calling to schedule a later time in the day for pickup.

Once he arrived at the pickup location, Bill collected the packages. It was not uncommon for the customer to have several packages for delivery. In addition to the name and address of the delivery location, he also recorded the time of pickup. He noted the desired delivery time, the location of the delivery, and the weight of the package to determine the courier cost. When he picked up the package, he printed out a label with his portable printer that he kept in the delivery van.

At first, Bill required customers to pay at the time of pickup, but he soon discovered that there were some regular customers who preferred to receive a monthly bill for all their shipments. He wanted to be able to accommodate those customers. Bills were due and payable upon receipt.

To help keep track of all the packages, Bill decided that he needed to scan each package as it was sorted in the warehouse. This would enable him to keep good control of his packages and avoid loss or delays.

The delivery of a package was fairly simple. Upon delivery, he would record information about when the delivery was made and who received it. Because some of the packages were valuable, it was necessary in those instances to have someone sign for the package upon delivery.

Read the above information then answer the questions which follow.

1. From the above description, as well as the information from Chapter 2 (refer back to week 2 tutorial's case study), identify all the actors who will be using the system.

2. Using the actors who you identified in question 1, develop a list of use cases based on the <u>user goal technique</u>. Draw a use case diagram for these use cases.

3. Now using the <u>event decomposition technique</u> for each event you identify in the description here, name the event, state the type of event, and name the resulting use case.

Task 5

For this task, you will explore how to use ChatGPT *and* PerplexityAI:

1. Register for an account in ChatGPT by clicking the link above.
2. Use ChatGPT to answer questions 1 and 2 of the previous Task (task 4):
3. Discuss (your own words) <u>how effective</u> ChatGPT is in answering the questions, specifically:
   a. Question 1: identify the section in the case study where the actors are found.
   b. Question 2: identify the section in the case study where the use cases are found. Are the use cases it proposed, correct (can be computerised, satisfied Perfect Technology Assumptions and Elementary Business Process)?

4. Repeat the above steps using PerplexityAI. Which is better (ChatGPT or PerplexityAI)?